

Refresher

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space: outcomes $\omega \in \Omega$, events $A \in \mathcal{F}$, probabilities $\mathbb{P}(A)$.

1. **Random variable** $X: \Omega \rightarrow \mathbb{R}$ with $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$ for every x . *A number determined by the outcome.*
2. **Indicator** $\mathbf{1}_A: \Omega \rightarrow \{0, 1\}$, equal to 1 on A and 0 off A . *Records whether A occurred.*
3. **Distribution function** $F_X(x) = \mathbb{P}(X \leq x)$. *Probability accumulated up to x .*
4. **Probability mass function** $p_X(x) = \mathbb{P}(X = x)$. *Probability of each value.*
5. **Probability density function** $f_X: \mathbb{R} \rightarrow [0, \infty)$ with $\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$. *Probability is area under f_X .*
6. **Joint density** $f_{X,Y}: \mathbb{R}^2 \rightarrow [0, \infty)$ with $\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y} dy dx$. *Probability is volume under $f_{X,Y}$.*
7. **Expected value** $\mathbb{E}[X] = \sum_x x p_X(x)$ or $\int_{\mathbb{R}} x f_X(x) dx$. *Average of the values, weighted by probability.*
8. **Conditional probability** $\mathbb{P}(A | B) = \mathbb{P}(A \cap B) / \mathbb{P}(B)$. *Probability of A when B is known to occur.*
9. **Conditional density** $f_{X|Y}(x | y) = f_{X,Y}(x, y) / f_Y(y)$. *Density of X when $Y = y$ is known.*

Random Variables

Definition

A **random variable** is a function $X: \Omega \rightarrow \mathbb{R}$ such that $\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$.

X is a number determined by the outcome ω ; the condition says that “ $X \leq x$ ” is an event, so it has a probability.

Notation

For a statement about X , we write the event and its probability without ω :

$$\{X \in A\} := \{\omega \in \Omega : X(\omega) \in A\}, \quad \mathbb{P}(X \in A) := \mathbb{P}(\{\omega \in \Omega : X(\omega) \in A\}),$$

and likewise $\mathbb{P}(X = x)$, $\mathbb{P}(a \leq X \leq b)$, $\mathbb{P}(X \leq x)$.

Example

Two dice: $\Omega = \{1, \dots, 6\}^2$, each outcome having probability $\frac{1}{36}$, and $X(\omega) = \omega_1 + \omega_2$. Then $\{X = 7\} = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}$, so $\mathbb{P}(X = 7) = \frac{6}{36}$.

Indicator Functions

Definition

For an event $A \in \mathcal{F}$, the **indicator** of A is the random variable $\mathbf{1}_A: \Omega \rightarrow \{0, 1\}$,

$$\mathbf{1}_A(\omega) = \begin{cases} 1 & \omega \in A, \\ 0 & \omega \notin A. \end{cases}$$

$\mathbf{1}_A$ records as a number whether A occurs: it equals 1 when A occurs and 0 otherwise, so $\mathbb{P}(\mathbf{1}_A = 1) = \mathbb{P}(A)$.

- $\mathbf{1}_{A \cap B} = \mathbf{1}_A \mathbf{1}_B$, $\mathbf{1}_{A^c} = 1 - \mathbf{1}_A$, $\mathbf{1}_{A \cup B} = \mathbf{1}_A + \mathbf{1}_B - \mathbf{1}_A \mathbf{1}_B$.
- If X is the number of the events $A_1, \dots, A_n$ that occur, then $X = \mathbf{1}_{A_1} + \dots + \mathbf{1}_{A_n}$.

Distribution Function (cdf)

Definition

The **cumulative distribution function** of X is $F_X: \mathbb{R} \rightarrow [0, 1]$,

$$F_X(x) = \mathbb{P}(X \leq x).$$

$F_X(x)$ is the probability that X is at most x ; it exists for every random variable, discrete or continuous.

- F_X is non-decreasing, $F_X(x) \rightarrow 0$ as $x \rightarrow -\infty$ and $F_X(x) \rightarrow 1$ as $x \rightarrow \infty$.
- $\mathbb{P}(a < X \leq b) = F_X(b) - F_X(a)$.
- F_X determines every probability $\mathbb{P}(X \in A)$. The pmf and the pdf are two convenient ways of describing it.

Probability Mass Function (pmf)

Definition

Let X take countably many values. The **pmf** of X is $p_X: \mathbb{R} \rightarrow [0, 1]$,

$$p_X(x) = \mathbb{P}(X = x).$$

p_X gives the probability of each possible value, and every probability $\mathbb{P}(X \in A)$ is a sum of these.

- $\sum_x p_X(x) = 1$, the sum running over the values of X .
- $\mathbb{P}(X \in A) = \sum_{x \in A} p_X(x)$; in particular $F_X(x) = \sum_{t \leq x} p_X(t)$.

Probability Density Function (pdf)

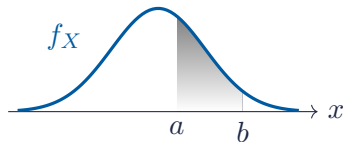
Definition

A **pdf** of X is a function $f_X: \mathbb{R} \rightarrow [0, \infty)$ such that

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) \, dx \quad \text{for all } a \leq b.$$

Probability is area under f_X ; the height $f_X(x)$ is probability per unit length near x , not a probability.

- $\int_{-\infty}^{\infty} f_X(x) \, dx = 1$, $F_X(x) = \int_{-\infty}^x f_X(t) \, dt$,
 $f_X = F'_X$ where F_X is differentiable.
- $\mathbb{P}(X = x) = \int_x^x f_X = 0$ for every x , and $f_X(x)$ may exceed 1; the uniform distribution on $[0, \frac{1}{2}]$ has $f_X = 2$ there.



Shaded area = $\mathbb{P}(a \leq X \leq b)$.

Joint Probability Density Function

Definition

A **joint pdf** of X and Y is a function $f_{X,Y}: \mathbb{R}^2 \rightarrow [0, \infty)$ such that, for all $a \leq b$ and $c \leq d$,

$$\mathbb{P}(a \leq X \leq b, c \leq Y \leq d) = \int_a^b \int_c^d f_{X,Y}(x, y) dy dx.$$

(X, Y) is a random point in the plane; probability is volume under the surface $f_{X,Y}$, and $\iint_{\mathbb{R}^2} f_{X,Y} = 1$.

- **Marginals.** Integrating out one variable leaves the pdf of the other:
 $f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y) dy$, $f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y) dx$.
- **Independence.** $X \perp\!\!\!\perp Y$ means $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$ for all A, B ;
equivalently $f_{X,Y}(x, y) = f_X(x)f_Y(y)$.

Example: breaking a stick twice. The marginal density of X is unbounded.

Break a unit stick at a uniform point Y , then the left piece at a uniform point X . Then $f_{X,Y}(x, y) = \frac{1}{y}$ for $0 < x < y < 1$, $f_Y(y) = 1$, and $f_X(x) = \int_x^1 \frac{dy}{y} = -\ln x$.

Discrete and Continuous Side by Side

	X with pmf p_X	X with pdf f_X
$\mathbb{P}(a \leq X \leq b)$	$\sum_{a \leq x \leq b} p_X(x)$	$\int_a^b f_X(x) \, dx$
$F_X(x) = \mathbb{P}(X \leq x)$	$\sum_{t \leq x} p_X(t)$	$\int_{-\infty}^x f_X(t) \, dt$
total mass	$\sum_x p_X(x) = 1$	$\int_{\mathbb{R}} f_X(x) \, dx = 1$
$\mathbb{E}[X]$	$\sum_x x p_X(x)$	$\int_{\mathbb{R}} x f_X(x) \, dx$
$\mathbb{E}[g(X)]$	$\sum_x g(x) p_X(x)$	$\int_{\mathbb{R}} g(x) f_X(x) \, dx$

Expected Value

Definition

$$\mathbb{E}[X] = \sum_x x p_X(x) \text{ if } X \text{ has pmf } p_X, \quad \mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \text{ if } X \text{ has pdf } f_X,$$

$\mathbb{E}[X]$ is the average of the possible values of X , each weighted by its probability.

- $\mathbb{E}[g(X)] = \sum_x g(x) p_X(x)$, respectively $\int_{\mathbb{R}} g(x) f_X(x) dx$.
- Linearity: $\mathbb{E}[aX + bY] = a \mathbb{E}[X] + b \mathbb{E}[Y]$ for any random variables X, Y .
- Indicators: $\mathbb{E}[\mathbf{1}_A] = 1 \cdot \mathbb{P}(\mathbf{1}_A = 1) + 0 \cdot \mathbb{P}(\mathbf{1}_A = 0) = \mathbb{P}(A)$.

Example: the hat-check problem

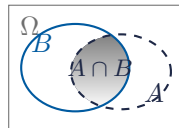
n hats are returned to their n owners in uniformly random order. With $A_i = \{\text{person } i \text{ gets their own hat}\}$, $\mathbb{P}(A_i) = \frac{1}{n}$, and the number of correct returns is $X = \sum_i \mathbf{1}_{A_i}$, so $\mathbb{E}[X] = \sum_i \mathbb{P}(A_i) = n \cdot \frac{1}{n} = 1$ for every n .

Conditional Probability

Definition

For events $A, B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$



The probability of A when B is known to occur.

- Conditional probabilities obey the usual rules, e.g.
 $\mathbb{P}(A^c \mid B) = 1 - \mathbb{P}(A \mid B)$.
- $\mathbb{P}(A \cap B) = \mathbb{P}(A \mid B) \mathbb{P}(B)$.
- $A \perp\!\!\!\perp B$ iff $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ iff $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Example: a rare disease

1% of people have disease D . A test T is positive for 99% of the sick and 5% of the healthy. Then $\mathbb{P}(D \mid T) = \mathbb{P}(D \cap T) / \mathbb{P}(T) = 0.01 \cdot 0.99 / (0.01 \cdot 0.99 + 0.99 \cdot 0.05) \approx 0.17$.

Conditional Density

Definition

Let X, Y have joint pdf $f_{X,Y}$ and $S_Y = \{y \in \mathbb{R} : f_Y(y) > 0\}$. The **conditional pdf** of X given Y is the function

$$f_{X|Y} : \mathbb{R} \times S_Y \rightarrow [0, \infty), \quad f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

The density of X when $Y = y$ is known: slice the surface $f_{X,Y}$ along the line $Y = y$ and rescale the slice to area 1.

- The two arguments play different roles: y is the *observed* value, restricted to where $f_Y(y) > 0$, and x is the variable of the resulting density.
- For each fixed $y \in S_Y$, the function $x \mapsto f_{X|Y}(x | y)$, $\mathbb{R} \rightarrow [0, \infty)$, is a pdf: it is ≥ 0 and integrates to $f_Y(y)/f_Y(y) = 1$.

Conditional Density: Discrete Version and Example

Definition (discrete version)

Let X, Y have joint pmf $p_{X,Y}$ and $S_Y = \{y : p_Y(y) > 0\}$. The **conditional pmf** of X given Y is the function

$$p_{X|Y} : \mathbb{R} \times S_Y \rightarrow [0, 1], \quad p_{X|Y}(x | y) = \frac{p_{X,Y}(x, y)}{p_Y(y)} = \mathbb{P}(X = x | Y = y).$$

Each value is an ordinary conditional probability, of the event $\{X = x\}$ given the event $\{Y = y\}$. For fixed y , $x \mapsto p_{X|Y}(x | y)$ is a pmf.

Example: two dice

Y the first die, X the sum, $S_Y = \{1, \dots, 6\}$. For $y \in S_Y$: $p_{X|Y}(x | y) = \frac{1/36}{1/6} = \frac{1}{6}$ if $x \in \{y + 1, \dots, y + 6\}$, else 0.

Example: the stick, continued

$S_Y = (0, 1)$; for $y \in S_Y$, $f_{X|Y}(x | y) = (1/y)/1 = 1/y$ if $0 < x < y$, else 0: given the first break at y , the second is uniform on $[0, y]$.